Poster: Towards Transparent NIDS: An xLSTM-Based Explainable Architecture

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2026 Silicon Valley Cybersecurity Conference (SVCC) | 979-8-3195-0664-1/26/$31.00 ©2026 IEEE | DOI: 10.1109/SVCC69905.2026.11642322

***Abstract*—Current efforts in AI-based Network Intrusion Detection Systems (NIDSs) have predominantly explored the use of deep learning-based models. Although these methods demonstrate superior performance, the internal mechanism of these models remains opaque. Additionally, existing efforts in explainable NIDS primarily rely on post-hoc techniques like LIME and SHAP, which lack robustness and fail to consider feature-level interactions in fast-paced environments. To address these limitations, we propose an explainable NIDS framework based on the Extended Long Short-Term Memory (xLSTM) architecture. By leveraging the xLSTM network, this framework achieves intrinsic interpretability without sacrificing detection accuracy. Preliminary evaluations on the UNSW-NB15 dataset demonstrate promising performance and a strong foundation for a transparent and robust application for network security.**

***Index Terms*—Network Intrusion Detection Systems (NIDS), Extended Long Short-Term Memory (xLSTM), Explainable AI (XAI), Deep Learning.**

I. Introduction

The escalating sophistication of cyber threats necessitates robust Network Intrusion Detection Systems (NIDSs). Although deep learning has significantly advanced NIDS capabilities, the field struggles with a critical trade-off between model performance and explainability [1] [2].

Historically, Recurrent Neural Networks (RNNs) and Long Short-Term Memory (LSTM) networks are highly effective at capturing temporal dependencies in network traffic. However, they suffered from limitations in parallelization and capturing long-range dependencies [3]. Recently, transformer architectures have dominated the landscape due to their powerful self-attention mechanisms. However, transformers are probabilistic models that struggle with large contexts and operating in real-time [4]. In the context of NIDS, where millions of packets must be processed to make rapid decisions, such limitations reduce the utility in real-world environments.

Furthermore, the deployment of deep learning in NIDS is hindered by the black-box nature of these models. In cybersecurity, transparency is paramount to security analysts who must understand *why* an anomaly was flagged to trust the NIDS and respond effectively. Prior efforts in eXplainable AI (XAI) for NIDS have predominantly relied on post-hoc perturbation techniques, such as LIME and SHAP [5]. These methods are not robust against varying adversarial inputs, and their added computational cost makes them unsuitable for real-time network environments.

To address these challenges, we introduce the application of the Extended Long Short-Term Memory (xLSTM) [6] architecture to NIDS [7]. xLSTM provides performance comparable to transformers but maintains linear time computational complexity and significantly lowers operational cost. This research aims to utilize xLSTM's unique architectural features to achieve intrinsic interpretability without sacrificing detection accuracy.

II. Proposed Methodology

This research proposes a novel explainable NIDS framework that shifts away from post-hoc approximations to the inherent structural interpretability of xLSTM architecture.

A. xLSTM for Network Traffic Analysis

Traditional LSTMs use sigmoid activation functions for their input and forget gates, which restricts the values between 0 and 1. This limits the network's ability to adapt its memory when it encounters highly relevant information. Moreover, LSTMs are constrained by a scalar memory cell state, which creates a bottleneck that forces all information to be heavily compressed.

xLSTM extends traditional LSTM capabilities by introducing exponential gating and matrix memory structures. xLSTM is constructed using two cell variants, namely sLSTM and mLSTM. Exponential gating allows the network to selectively emphasize or suppress memory updates in real-time. The matrix memory increases the network's capacity, allowing it to retain high-dimensional relationships among various network events.

B. Explainability Framework

Rather than treating the model as a black box, we propose a comprehensive framework that leverages xLSTM's internal mechanisms for explainability. We are currently testing interpretability methodologies on the xLSTM architecture, such as gate activation monitoring. Visualizing the xLSTM gating mechanism provides insight into specific network event information the model decides to retain or discard.

III. Experimental Design

To validate the proposed xLSTM framework, we conducted evaluations on standard cybersecurity benchmarks and compared its results with the baseline LSTM model.

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A. Datasets

The UNSW-NB15 [8] and TON-IoT [9] network intrusion datasets are used for preliminary evaluations.

B. Evaluation Metrics

The framework is evaluated on two distinct fronts:

1) Detection Efficacy: Standard metrics, including accuracy, precision, recall, F1 score, and false positive rate, will be compared against baseline transformer and traditional LSTM models.
2) Explainability: Interpreting gate activations in xLSTM to understand the model's behavior when subjected to normal versus malicious network traffic.

C. Environmental Configuration

The model training and preliminary evaluations were executed on the Apple M5 device with a 10-core CPU, a 10-core GPU, and 16 GB of unified memory. The xLSTM architecture and subsequent data preprocessing pipelines were implemented using Python and PyTorch.

IV. Results

We evaluated xLSTM comprising two sequential blocks. The base block utilizes an mLSTM layer configured with 4 heads and a 1D convolution kernel of size 4, and the secondary block utilizes an sLSTM layer, also configured with 4 heads and a kernel size of 4. For comparison, we also evaluated a vanilla LSTM model with two sequential blocks and embedding dimensions of 54. These configurations were selected to roughly match the number of parameters in both models.

A. Detection Efficiency

TABLE I
BINARY CLASSIFICATION RESULTS OF LSTM AND XLSTM ON THE UNSW-NB15 DATASET.

Model / Class	Precision	Recall	F1-Score	Occurrences
LSTM				
Normal	0.93	0.91	0.92	55,951
Attack	0.96	0.97	0.96	119,340
Macro Avg	0.94	0.94	0.94	175,291
Weighted Avg	0.95	0.95	0.95	175,291
Accuracy			0.95	175,291
xLSTM				
Normal	0.95	0.91	0.93	55,951
Attack	0.96	0.98	0.97	119,340
Macro Avg	0.95	0.94	0.95	175,291
Weighted Avg	0.95	0.95	0.95	175,291
Accuracy			0.95	175,291

The xLSTM architecture outperformed the baseline LSTM model on both datasets. On the UNSW-NB15 dataset, we evaluated xLSTM and LSTM on a binary classification problem. As shown in Table I, the xLSTM architecture achieved a test accuracy of 95.44% with a 2.25% false negative rate, and the LSTM achieved 94.91% accuracy with a 3.4% false negative rate. xLSTM exhibits a higher false positive rate, but minimizing false negatives is often prioritized in cybersecurity applications. Figures 1 and 2 show the confusion matrix for LSTM and xLSTM on UNSW-NB15.

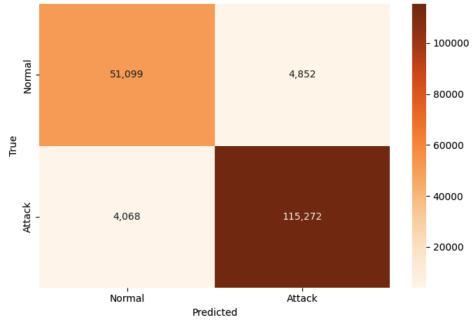


Fig. 1. Confusion Matrix of LSTM on the UNSW-NB15 Dataset.

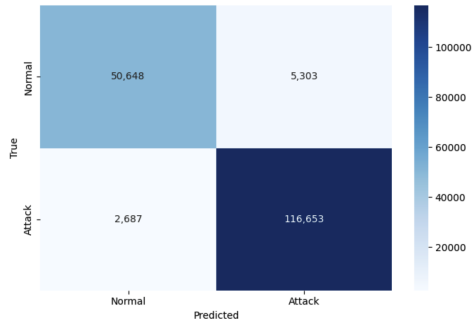


Fig. 2. Confusion Matrix of xLSTM on the UNSW-NB15 Dataset.

TABLE II
TON-IOT MULTI-CLASS DATASET DISTRIBUTION.

Label	Category	Train (80%)	Test (20%)	Percentage
0	Backdoor	162,898	40,724	20.4%
1	Normal	187,990	46,998	23.5%
2	Ransomware	58,244	14,561	7.3%
3	XSS	390,868	97,717	48.9%
Total		800,000	200,000	

We performed a multi-class evaluation on the TON-IoT dataset using a stratified split to maintain uniform representation of specific attack vectors and normal traffic, shown in Table II. This experiment accounts for class imbalance by incorporating class weights in the loss function during training.

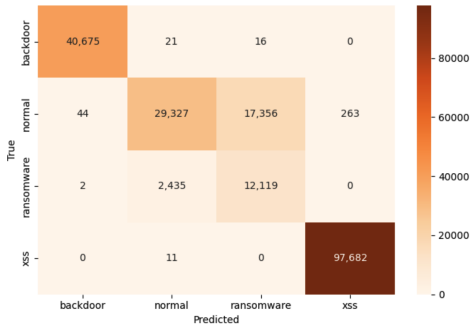


Fig. 3. Confusion Matrix of LSTM on the TON-IoT Dataset.

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TABLE III
MULTI-CLASS CLASSIFICATION RESULTS OF LSTM AND XLSTM ON THE TON-IOT DATASET.

Model / Class	Precision	Recall	F1-Score	Occurrences
LSTM				
Backdoor	1.00	1.00	1.00	40,712
Normal	0.92	0.62	0.74	46,990
Ransomware	0.41	0.83	0.55	14,556
XXS	1.00	1.00	1.00	97,693
Macro Avg	0.83	0.86	0.82	199,951
Weighted Avg	0.94	0.90	0.91	199,951
Accuracy			0.90	199,951
xLSTM				
Backdoor	1.00	1.00	1.00	40,712
Normal	0.94	0.86	0.90	46,990
Ransomware	0.65	0.84	0.73	14,556
XXS	1.00	1.00	1.00	97,693
Macro Avg	0.90	0.92	0.91	199,951
Weighted Avg	0.96	0.95	0.96	199,951
Accuracy			0.95	199,951

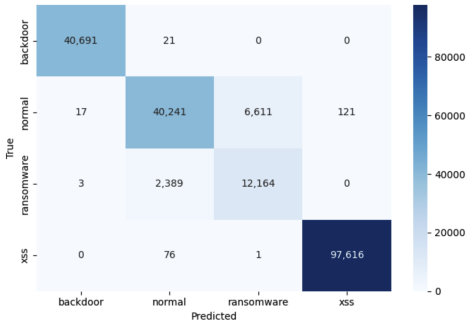


Fig. 4. Confusion Matrix of xLSTM on the TON-IoT Dataset.

As shown in Table III, xLSTM achieved a test accuracy of 95.38%, significantly higher than the LSTM (89.92%). Both models excel at detecting backdoor attacks but struggle in identifying ransomware attacks, likely due to the class imbalance in the dataset. Figures 3 and 4 show the confusion matrix for LSTM and xLSTM on TON-IoT.

The raw performance metrics are insufficient for security analysts to confidently rely on the model without being able to verify *why* the model made a specific prediction. To address this gap, we leverage xLSTM's internal architectural features for interpretability.

B. Explainability

Figure 5 shows how the information is accepted and discarded by the model's input gate for a given subset of network flow. For instance, across heads 1, 2, and 3, there is a suppression of input data between time steps 4 and 10. The model effectively determines that the network traffic during this window is irrelevant to the classification of the flow. On the other hand, the model shows maximum activation on time steps 1, 11, and 15. This indicates that the model is heavily weighing these network instances for its classification, likely indicating the beginning of a network event or a significant network episode.

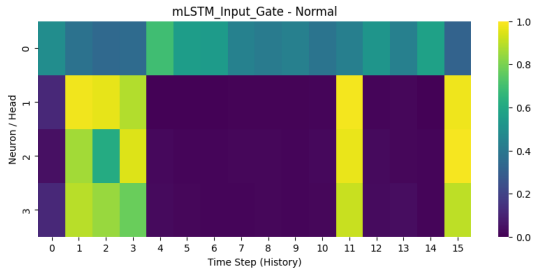


Fig. 5. mLSTM Input Gate Activation on Normal Network Traffic from UNSW-NB15.

V. CONCLUSION AND FUTURE WORK

This research proposes a novel explainable xLSTM framework for Network Intrusion Detection Systems (NIDSs). By shifting away from post-hoc perturbation-based techniques, we successfully utilize xLSTM's internal gating mechanisms to provide real-time visualizations to better interpret why specific network traffic is flagged. Future work includes evaluating the xLSTM architecture on other cybersecurity datasets, specifically NSL-KDD, CIC-IDS2017, and CIC-IDS2018. We will utilize these insights to propose architectural changes to xLSTM, tailoring it specifically for NIDS. We aim to utilize gate activations to help security analysts understand the model's behavior during classification. Moreover, this framework will be extended to generate deeper feature-level interactions, providing comprehensive semantic explanations for the model's outputs, and compared with other interpretability techniques such as SHAP and LIME.

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